

Subject Code		Name of the Subject								L	T	P	C			
BPCCS4020		Theory of Computation								4	0	1	4			
Course Educational Objectives																
CEO1	To introduce foundational concepts of alphabets, strings, formal languages, and grammars.															
CEO2	To enable students to design and analyze deterministic and non-deterministic finite automata (DFA & NFA).															
CEO3	To develop the ability to use regular expressions and grammars for recognizing regular languages and relate them with automata.															
CEO4	To explore context-free grammars, parsing algorithms, and pushdown automata as models for language recognition															
CEO5	To build understanding of Turing Machines, undecidability, and computational power limits.															
CEO6	To introduce recursive function theory and NP-completeness concepts to assess problem solvability and complexity.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Understand formal languages, grammars, and design of various finite automata.															
CO2	Apply regular expressions and finite automata in the context of language recognition and transformations.															
CO3	Design and analyze context-free grammars and pushdown automata for language parsing.															
CO4	Construct Turing machines and explore concepts of undecidability and context-sensitive languages.															
CO5	Evaluate problems using recursive function theory and understand the limits of computation through NP-completeness.															
CO6	Analyze and compare various automata models and understand their role in the Chomsky hierarchy.															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	3											3	3		
CO2	3		3	2									3	3		
CO3	3	3		3	3								3	3		
CO4	2		3		3								3	3		
CO5				3									3	3		
CO6		3											3	3		
SYLLABUS																
UNIT:1 (10 Hours)																
Mathematical preliminaries: Alphabet, String, Languages, Grammars, Strings and operations on strings.																
Finite Automata: Definition, Basic model, Types of Finite Automata, Design of DFA, Design of NFA, NFA vs. DFA, Eliminating ϵ -transitions from NFA, NFA to DFA conversion, NFA or DFA as a language acceptor, Minimization of Finite Automata. Equivalence of two Finite Automata																

UNIT:2 (12 Hours) Regular Expressions: Regular Set and Regular Expressions. Operators in Regular expressions, Building Finite Automata from Regular expression, Arden's theorem & Building Regular expression from Finite Automata, Pumping Lemma for Regular languages, Closure properties of Regular languages. Grammar: Definition, Regular Grammar, Regular Grammar to Finite Automaton, Finite Automaton to Regular Grammar
UNIT:3 (12 Hours) Grammar: Designing Context Free Grammar, String Derivation, Parse Tree Construction, Ambiguous Grammar, Chomsky and Greibach Normal Forms, CYK parsing algorithm, Closure Properties of CFL, Pumping Lemma for CFL, Introducing Non-Context Free Grammar, Chomsky Hierarchy. Push Down Automata: Basic Model, Components, Moves of a PDA, ID of a PDA, Design of Deterministic PDA and Non-deterministic PDA, PDA to CFG and CGA to PDA conversion.
UNIT:4 (11 Hours) Turing Machines: Basic Model, Components, move of a TM, ID of TM, Design of a TM, Variants Of Turing Machine, Recursively Enumerable Languages, Undecidable problems, Post correspondence problem as an Undecidable Problem. Linear Bounded Automata and Context Sensitive Languages
UNIT: 5 (9 Hours) Primitive Recursive functions: μ - Recursive functions, Cantor and Godel numbering, ackerman's function, Excursiveness of Ackermann and Turing computable functions. Church Turing hypothesis, Recursive and Recursively Enumerable sets, NP Completeness: P and NP, NP complete and NP Hard problems.
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: 1. <i>Introduction to the theory of computation, Michael Sipser, Cengage Learning.</i> 2. <i>Formal Language and Automata Theory, C. K. Nagpal, Oxford University Press.</i>
Reference Books: 1. <i>Theory of Computer Science : Automata, Languages And Computation by by K L P Mishra and N Chandrasekaran</i> 2. <i>Introduction to Automata Theory, Languages and Computation: J. E. Hopcroft, J.D Ullman, Pearson Education.</i>

